Analysis of the Conservation State of Ancient Books

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Abstract

This project delivers an integrated information system for analysing the conservation state of ancient books across a network of European libraries and archives. Heterogeneous data on environmental monitoring, storage sites, material composition, and production century are combined into a single analytic dataset. A Conservation Risk Index (CRI) supports prioritisation of restoration and preventive policies. Results are exposed through a reproducible Colab notebook and a Streamlit dashboard designed for non-technical decision-makers.

1. Problem Definition and Decision Context

1.1 Operational Objective

Curators and conservators must allocate limited resources to volumes at highest deterioration risk while maintaining preventive environmental control. The central question is:

Which books, in which sites, face elevated conservation risk given environment, materials, and age — and what interventions yield the largest risk reduction?

1.2 Conservator Decision Cycle

The system supports a five-stage cycle: monitor - assess - prioritise - intervene - re-monitor.

Stage	System support
Monitor	Temperature, RH, light, AQI, exceptional events per site
Assess	Compute CRI and risk level per volume
Prioritise	Rank volumes and sites; identify highest-risk interventions
Intervene	Restoration or environmental remediation
Re-monitor	Update campaign data and refresh indicators

1.3 Design Decisions

Choice	Rationale
Join on site_id	Environmental loggers are building-level in open heritage data, not per-volume
CRI scale 0-100	Interpretable by curators; directly usable for colour-coding on the map
Streamlit	Native Python integration with the pipeline; @st.cache_data for performance; free deployment on Streamlit Community Cloud
480 volumes	Sufficient for stable ML training and statistical tests; dashboard remains responsive

2. Information System Architecture

The system follows a layered BI architecture with four components:

Layer	Component	Technology
Data integration	ETL + join	src/data_integration.py -> integrated_heritage_books.csv
Analytics engine	CRI computation	src/conservation_index.py
Presentation	BI dashboard	app/streamlit_app.py -> Streamlit Community Cloud
Documentation	Reproducible analysis	notebooks/conservation_analysis.ipynb (Colab)

All layers share the same Python environment. The CRI formula is defined once in conservation_index.py and imported by both the notebook and the dashboard. One command — python -m src.data_integration — regenerates the full dataset from scratch with a fixed random seed (42), guaranteeing reproducibility.

3. Phase 1 — Dataset Retrieval and Construction

3.1 Sources

No single public dataset covered all required dimensions. Three source layers were integrated via a common site_id key:

Layer	Source	Licence
Sites	8 European institutions: Coimbra, Heidelberg, Bologna, Florence (x2), Paris (x2), London	Public institutional data
Environmental monitoring	Ferreira et al., Data in Brief (2019) — Coimbra historic library	CC BY
Volume metadata	Schema aligned with Europeana and Heidelberg UB open catalogues	CC BY-SA
Climate context	Koppen-style zone labels per city	Derived attribute

3.2 Integration

The three source tables are joined using pandas left-merge on site_id. Environmental monthly data is aggregated to a six-month campaign average per site before joining to volumes, producing integrated_heritage_books.csv (480 rows, 38 columns). Volume material distributions are conditioned on century — medieval books are 45% parchment; 19th-century books are 55% industrial paper, reflecting real historical patterns.

3.3 Data Quality and Limitations

Issue	Handling
Synthetic catalogue	480 volume records procedurally generated with realistic marginals. Not real Europeana IDs. Pipeline designed for API replacement.
Environmental series	Modelled from Ferreira et al. zone profiles with Gaussian noise and seasonal sine wave. Not a raw sensor download.
Condition ground truth	observed_condition_score (1-5) is a survey proxy; CRI is the primary decision index.
Missing values	Pipeline produces a complete analytic table; no imputation required.

4. Phase 2 — Analytical Methodology

4.1 Pre-processing

Type casting of boolean event flags; material label mapping to standard taxonomy (parchment, laid paper, industrial paper, mixed); outlier flagging for implausible RH/T values; missing value check.

4.2 Conservation Risk Index

The CRI is a composite score (0-100, higher = worse) inspired by UNI 10829 (thermo-hygrometric thresholds) and ICCROM preventive conservation guidelines. It combines four normalised stress components:

Component	Weight	Logic
Environmental stress	35%	Band stress on RH (45-55%), T (18-22C), light (50 lux max), AQI. Weighted: 0.4*RH + 0.3*T + 0.2*light + 0.1*AQI
Material stress	30%	Parchment=1.0, mixed=0.85, laid paper=0.75, industrial paper=0.45. Iron-gall ink=0.90
Age stress	25%	Years since production / 800, clipped to [0.1, 1.0]. Denominator spans 12th-20th century range.
Event stress	10%	Flood +0.50, HVAC failure +0.25, recent restoration -0.10. Capped at 1.

Risk labels: Low (CRI < 35) · Moderate (35-55) · High (55-75) · Critical (75+).

4.3 Analyses Performed

- ANOVA — one-way F-test of CRI across four material groups.
- Chi-square — contingency table of high_risk x flood_event.
- Random forest (200 trees, class_weight='balanced') — predicts elevated risk from 9 features; feature importance chart produced.
- K-means clustering (k=4, n_init=10) — groups volumes in four-dimensional stress space after StandardScaler normalisation.
- Geographic scatter plot of site mean CRI by coordinates.
- Monthly RH time series per site over the six-month campaign.

4.4 Key Results

- Mean CRI approximately 39 across 480 volumes; distribution is right-skewed with a small High/Critical tail.
- Bologna (Archiginnasio) shows the highest site mean CRI, driven by RH around 58-59% — above the 45-55% optimal band.
- Random forest: century and average humidity are the dominant predictors; ink type alone is a weaker signal.
- K-means identifies four profiles: old parchment in poor climate; old parchment in stable climate; modern paper in any climate; mixed event-exposed volumes.

Operational implication: stabilise RH before scheduling restoration on old volumes in poor microclimates. The what-if tab quantifies the CRI reduction from a 5% RH improvement.

5. Phase 3 — Business Intelligence Dashboard

5.1 Framework Choice

Streamlit was chosen for three concrete reasons. First, it shares the same Python environment as the data pipeline — the dashboard imports directly from src/conservation_index.py without any serialisation layer. Second, @st.cache_data keeps the dataset in memory across filter interactions. Third, free deployment on Streamlit Community Cloud required no infrastructure setup. Trade-off: Power BI would offer richer enterprise visuals and role-based access control but would require decoupling the CRI logic from Python entirely.

5.2 Dashboard Functionality

Component	Functionality
KPI row	Volumes analysed · Mean CRI · High/Critical count · Sites above threshold
Sidebar filters	Century · Material · Country · Site type · RH range · Temperature range
Overview tab	CRI histogram · Mean CRI by material · CRI by century boxplot · Correlation matrix
Geography & time	Plotly mapbox site map · Monthly RH/T time series · CRI by century lines
Drill-down	Site to volume table to individual volume JSON detail (materials, environment, events)
What-if & priorities	Environmental sliders to simulated CRI delta · Top-N priority intervention ranking

Deployment: Streamlit Community Cloud (app/streamlit_app.py). Public URL in README.md.

6. Critical Self-Assessment

6.1 Strengths

- End-to-end reproducible pipeline — one command regenerates the full dataset from scratch.
- CRI grounded in UNI 10829 and conservation literature; weights are documented and motivated.
- Honest data lineage — synthetic catalogue is explicitly acknowledged throughout.
- Dashboard covers all required BI features: KPIs, filters, map, drill-down, decision tool.

6.2 Weaknesses and Future Work

- Replace synthetic volumes with real Europeana API records (TYPE:TEXT filter).
- Ingest raw Coimbra CSV from Ferreira et al. supplementary data for site S01.
- Build volume-level microclimate model — site averages may underestimate risk near exterior walls.
- Validate CRI weights against a labelled conservator panel dataset.
- Add air quality sensor data where available (ICCU, national archive networks).

6.3 Recognised Limitations

The CRI weights (35/30/25/10) are expert-motivated but not statistically calibrated against ground-truth deterioration outcomes. Site-level environment applied uniformly to all stacks may underestimate risk for volumes near exterior walls. The six-month monitoring window may not capture seasonal extremes.

7. Conclusion

This project demonstrates how heterogeneous open heritage data can be integrated into a decision-support system for book conservation. The CRI provides curators with an actionable ranking of volumes by deterioration risk, grounded in environmental science and material knowledge. The Streamlit dashboard allows a non-technical conservator to identify at-risk volumes, understand the drivers, and simulate environmental interventions before committing resources. Limitations are explicit and the architecture is designed for incremental replacement of synthetic data with real institutional sources.

References

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